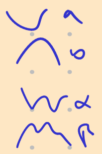
System

$$H_S = E_{MF} + \sum_k \psi_k^\dagger \begin{pmatrix} \eta_k + \xi_k & -\Delta \\ -\Delta & \eta_k - \xi_k \end{pmatrix} \psi_k \qquad \psi_k = \begin{pmatrix} c_{a,k} \\ c_{b,k} \end{pmatrix} \quad (\text{prej } a_k, b_k)$$



$$E_{\alpha/\beta,k} = \eta_k \pm \sqrt{\xi_k^2 + \Delta^2}$$

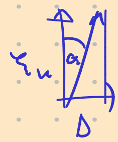
$$\left.\begin{aligned} \eta_k &= \frac{\bar{\varepsilon}_a + \bar{\varepsilon}_b}{2} \\ \xi_k &= \frac{\bar{\varepsilon}_a - \bar{\varepsilon}_b}{2} \end{aligned}\right\} \quad \begin{aligned} \eta_k + \xi_k &= \bar{\varepsilon}_a = \varepsilon_a(k) + V n_b \\ \eta_k - \xi_k &= \bar{\varepsilon}_b = \varepsilon_b(k) + V n_a \end{aligned}$$

$$\Delta = \sum_k \frac{V \cdot \Delta}{2\sqrt{\xi_k^2 + \Delta^2}} (n_{\beta,k} - n_{\alpha,k})$$

$$\Delta = \sum_k m_k \langle c_{a,k}^\dagger c_{b,k} \rangle \qquad \gamma_{\lambda,k} = \begin{pmatrix} \alpha_k \\ \beta_k \end{pmatrix} \begin{matrix} \lambda = \alpha \\ \lambda = \beta \end{matrix}$$

$$\psi_k = \begin{pmatrix} u_k & v_k \\ -v_k & u_k \end{pmatrix} \gamma_k$$

$$\rightarrow \quad \begin{matrix} u_k^2 \\ v_k^2 \end{matrix} = \frac{1}{2}\left(1 \pm \frac{\xi_k}{\sqrt{\xi_k^2 + \Delta^2}}\right)$$



$$\rightarrow \quad u_k v_k = \frac{1}{2} \sin(\theta_k) = \frac{\Delta}{2\sqrt{\xi_k^2 + \Delta^2}}$$

$$\underline{H_S = E_{MF} + \sum_k \left(E_{\alpha,k} n_{\alpha,k} + E_{\beta,k} n_{\beta,k}\right)} \qquad n_{\lambda,k} = \langle \gamma_{\lambda k}^\dagger \gamma_{\lambda k} \rangle_\rho$$

Bath:

$$H_B = \sum_{\lambda, q} \omega_{\lambda, q} a_{\lambda q}^\dagger a_{\lambda q}, \quad \omega_{\lambda, q} = \sqrt{\omega_{\lambda,0}^2 + c^2 |\vec{q}|^2} \qquad (\text{na primer})$$

Interakcija:

od začetka! N realni reprezentacij: $\lambda = \{a, b\}$ $\qquad \phi = \langle c_{a,j}^\dagger c_{b,j} \rangle$

$$H = \sum_{i,j,\lambda} \left(\varepsilon^{(\lambda)} \delta_{ij} + t_{ij}^{(\lambda)} e^{-ig(a_i^\dagger + a_i)}\right) c_{\lambda,i}^\dagger c_{\lambda,j} + h.c. +$$

$$+ V n_b \sum_j c_{a,j}^\dagger c_{a,j} + V n_a \sum_j c_{b,j}^\dagger c_{b,j} - V\phi \sum_j c_{b,j}^\dagger c_{a,j} - V\phi^* \sum_j c_{a,j}^\dagger c_{b,j} + E_{MF}$$

$$\rightarrow \quad c_{\lambda,j}^\dagger = \frac{1}{\sqrt{N}} \sum_k e^{ikR_j} c_{\lambda,k}^\dagger \quad \longrightarrow \quad n_\lambda = \frac{1}{N} \sum_{k,p} e^{i(k-p)R_j} \langle c_{\lambda,k}^\dagger c_{\lambda,p} \rangle = \frac{1}{N} \sum_k \langle c_{\lambda,k}^\dagger c_{\lambda,k} \rangle$$

$$\Delta \equiv + \frac{V}{N} \sum_k \langle c_{a,k}^\dagger c_{b,k} \rangle$$

$$\rightarrow \quad e^{-ig(a_i^\dagger + a_i)} = e^{-iga_i^\dagger} e^{-iga_i} e^{\overbrace{[a_i^\dagger, a_i]}^{-1} g^2/2} \simeq (1 - iga_i^\dagger)(1 - iga_i) \simeq \underline{\underline{1 - ig(a_i^\dagger + a_i)}}$$

$$e^{[a_i^\dagger, a_i] g^2/2} \simeq 1$$

$$\bullet \quad \sum_{k,p} \frac{1}{N} e^{i(k-p)R_j} \sum_{j,\lambda} \left(\varepsilon^{(\lambda)} + t^{(\lambda)}(e^{ik} + e^{-ik})\right) c_{\lambda,k}^\dagger c_{\lambda,p} = \sum_{k,\lambda} \left(\varepsilon^{(\lambda)} + 2t^{(\lambda)} \cos(k)\right) c_{\lambda,k}^\dagger c_{\lambda,k}$$

s.b. interakcija :

$$a_j^\dagger = \frac{1}{\sqrt{N}} \sum_q e^{iqR_j} a_q^\dagger$$

$$\to -ig \sum_{j,\lambda} t^{(\lambda)} \left(\frac{1}{\sqrt{N}} \sum_q \left(e^{iqR_j} a_q^\dagger + e^{-iqR_j} a_q \right)\right) \left[\frac{1}{N} \sum_{k,p} e^{i(k-p)R_j} e^{ik} c_{\lambda,k}^\dagger c_{\lambda,p} \right] + h.c. =$$

$$\frac{1}{N} \sum_j e^{i(k-p\pm q)} = \delta_{p,k\pm q}$$

$$= -i\frac{g}{\sqrt{N}} \sum_{\lambda,k,q} a_q^\dagger \quad c_{\lambda,k}^\dagger c_{\lambda,k+q}\, t^{(\lambda)} \left(e^{ik} - \underbrace{e^{i(k+q)}}_{h.c.} \right) -$$

$$-i\frac{g}{\sqrt{N}} \sum_{\lambda,k,q} a_q\, c_{\lambda,k}^\dagger c_{\lambda,k-q}\, t^{(\lambda)} \left(e^{ik} - \underbrace{e^{i(k-q)}}_{h.c.} \right) =$$

$q \to -q$

$$= \sum_{\lambda,k,q} \left(a_q^\dagger + a_{-q} \right) \underbrace{\left[-i\frac{g}{\sqrt{N}} t^{(\lambda)} \left(e^{ik} - e^{i(k+q)} \right) \right]}_{G_\lambda(k,q)} c_{\lambda,k}^\dagger c_{\lambda,k+q}$$

band index

diagonalno v λ : $c_a^\dagger c_a$ ali $c_b^\dagger c_b$

$$\to H_{sB} = \sum_{\lambda,k,q} \left(a_q^\dagger + a_{-q} \right) \underbrace{G_\lambda(k,q)\, c_{\lambda,k}^\dagger c_{\lambda,k+q}}$$

rotacija $\to$ $c_{\lambda,k} = U_{\mu\lambda,k}\, \gamma_{\mu,k}$ $\quad \begin{pmatrix} c_{a,k} \\ c_{b,k} \end{pmatrix} = \begin{pmatrix} u & v \\ -v & u \end{pmatrix} \begin{pmatrix} \alpha_k \\ \beta_k \end{pmatrix}$

$$\to \underbrace{G_\lambda(k,q)\, U_{\mu\lambda}(k)\, U_{\nu\lambda}(k+q)}_{g_{\mu\nu}(k,q)}\, \gamma_{\mu,k}^\dagger \gamma_{\nu,k+q}$$

$' = k+q$

$$\to \frac{\lambda = a:}{\lambda = b} \quad c_{a,k}^\dagger c_{a,k+q} = \left(u\alpha^\dagger + v\beta^\dagger \right)\left(u'\alpha' + v'\beta' \right) = uu'\left(\alpha^\dagger \alpha'\right) + uv'\left(\alpha^\dagger \beta'\right) + vu'\left(\beta^\dagger \alpha'\right) + vv'\left(\beta^\dagger \beta'\right)$$

(for $\lambda = b$: $u \to -v$, $v \to u$, $u' \to -v'$, $v' \to u'$; $uu' \to vv'$, $uv' \to -vu'$, $vu' \to -uv'$, $vv' \to uu'$)

$$\to g_{\alpha\alpha}(k,q) = u_k u_{k+q} G_a(k,q) + v_k v_{k+q} G_b(k,q) \qquad \alpha(k) \xrightarrow[\gamma(q)]{} \alpha(k+q)$$

$$\to g_{\alpha\beta}(k,q) = u_k v_{k+q} G_a(k,q) - v_k u_{k+q} G_b(k,q) \qquad \beta \longrightarrow \alpha$$

$$\to g_{\beta\alpha}(k,q) = v_k u_{k+q} G_a(k,q) - u_k v_{k+q} G_b(k,q) \qquad \alpha \longrightarrow \beta$$

$$\to g_{\beta\beta}(k,q) = v_k v_{k+q} G_a(k,q) + u_k u_{k+q} G_b(k,q) \qquad \beta \longrightarrow \beta$$

$$\to H_{sB} = \sum_{qk} \left(a_q^\dagger + a_{-q} \right) \left[g_{\alpha\alpha}\, \alpha_k^\dagger \alpha_{k+q} + g_{\alpha\beta}\, \alpha_k^\dagger \beta_{k+q} + g_{\beta\alpha}\, \beta_k^\dagger \alpha_{k+q} + g_{\beta\beta}\, \beta_k^\dagger \beta_{k+q} \right]$$

→ Lindblad: $L^{(\mu\nu)}_{kq} \propto \mu^{+}_{k} \nu_{k+q}$, $\{\mu,\nu\} = \{\alpha,\beta\}$.

→ rates? $\gamma_{fi} = 2\pi \left|\langle f | H_{SO} | i \rangle\right|^2 \delta(E_f - E_i)$

$$H_{SO} = \sum_{\substack{kq\\ \mu\nu}} \left(a_q^{+} g_{\mu\nu}(k,q)\, \gamma^{+}_{\mu,k} \gamma_{\nu,k+q} + a_q\, g_{\mu\nu}(k+q,-q)\, \gamma^{+}_{\mu,k+q} \gamma_{\nu,k} \right)$$

(1) ABSORPCIJA fotona: $|i\rangle = |N_q\rangle \otimes |1_{\nu,k}\, 0_{\mu,k+q}\rangle$

$|f\rangle = |N_q - 1\rangle \otimes |0_{\nu,k}\, 1_{\mu,k+q}\rangle$

$$\underbrace{\langle N_q - 1 | a_q | N_q \rangle}_{\sqrt{N_q}} \langle 0_{\nu,k} 1_{\mu,k+q} | \underbrace{\gamma^{+}_{\mu,k+q} \gamma_{\nu,k} | 1_{\nu,k} 0_{\mu,k+q} \rangle}_{|0_{\nu,k} 1_{\mu,k+q}\rangle} g_{\mu\nu}(k+q,-q)$$

$$M_1^{\mu\nu} = \sqrt{N_q} \cdot g_{\mu\nu}(k+q,-q)$$

(2) emisija $k+q \to k$: $|i\rangle = |N_q\rangle |0_{\mu,k} 1_{\nu,k+q}\rangle$

$|f\rangle = |N_q + 1\rangle |1_{\mu,k} 0_{\nu,k+q}\rangle$

$$\underbrace{\langle N+1 | a^{+} | N \rangle}_{\sqrt{N_q+1}} \langle 1_{\mu,k} 0_{\nu,k+q} | \gamma^{+}_{\mu,k} \gamma_{\nu,k+q} | 0_{\mu,k} 1_{\nu,k+q} \rangle g_{\mu\nu}(k,q) = M_2^{\mu\nu}$$

?

abs

em

$\alpha\alpha$:
$$\gamma^{\alpha\alpha}_{kq} = 2\pi \left| g_{\alpha\alpha}(k+q,-q) \right|^2 N_q \left(n_{\alpha,k}(1 - n_{\alpha,k+q}) \right) \delta(E_{\alpha,k+q} - E_{\alpha,k} - \omega_q) + $$
$$+ 2\pi \left| g_{\alpha\alpha}(k,q) \right|^2 (N_q+1) \left(n_{\alpha,k+q}(1 - n_{\alpha,k}) \right) \delta(E_{\alpha,k} - E_{\alpha,k+q} + \omega_q)$$

$\beta\beta$ isto!

$\alpha\beta$:
$$\gamma^{\alpha\beta}_{kq} = 2\pi \left| g_{\alpha\beta}(k+q,-q) \right|^2 N_q \left(n_{\beta,k}(1 - n_{\alpha,k+q}) \right) \delta(E_{\alpha,k+q} - \dots) +$$
$$+ 2\pi \left| g_{\alpha\beta}(k,q) \right|^2 (N_q+1) \left(n_{\alpha,k}(1 - n_{\beta,k+q}) \right) \delta(\dots)$$

?